LUNAR IMPACT AND PREDICTIVE ANALYSIS SYSTEM

Building and Evaluating a Cross Hazard Machine Learning Approach to Unified Lunar Forecasting and Mission Planning

Luka Takki · Luxembourg 2026

Abstract

LIPAS (Lunar Impact and Predictive Analysis System) is a hybrid forecasting and decision-support platform that combines Long Short-Term Memory (LSTM) machine learning with physics-based models to predict lunar space weather (temperature, radiation, micrometeorites, and solar storms). Users can view forecasts for any lunar location on an interactive web dashboard. This study evaluated whether a system like LIPAS could be developed, whether it was usable, and whether it could produce accurate forecasts. Results: ∼15% lower solar RMSE and ∼22% lower radiation RMSE compared with NASA-replicated flagship models, with earlier solar storm EVA caution alerts than NASA (T+45 vs T+55).

Introduction

Renewed lunar exploration through initiatives such as NASA's Artemis programme increases the need for faster and more accurate operational forecasting tools. Existing models often address individual hazards, requiring mission planners to consult multiple products when making time- critical decisions. LIPAS addresses this challenge by integrating forecasting approaches into a single decision-support platform enhanced by machine learning.

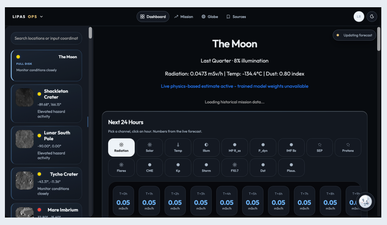
This study addresses three research questions:

RQ1: Can physics and machine learning be combined into an hybrid lunar forecast system?

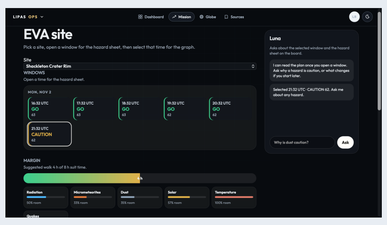
RQ2: Does unifying multiple hazards into one mission platform improve operational usability?

RQ3: Can LIPAS improve selected hazard forecasts versus established NASA/NOAA baselines?

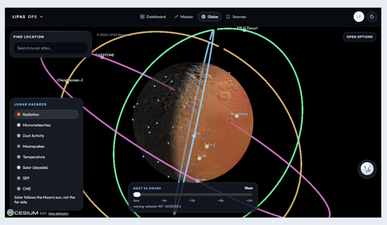
System Overview



Dashboard
Live six hazard forecasts, maps, trends, and alerts in one view.



Mission planner
Go / caution / no go for an EVA from place and duration.



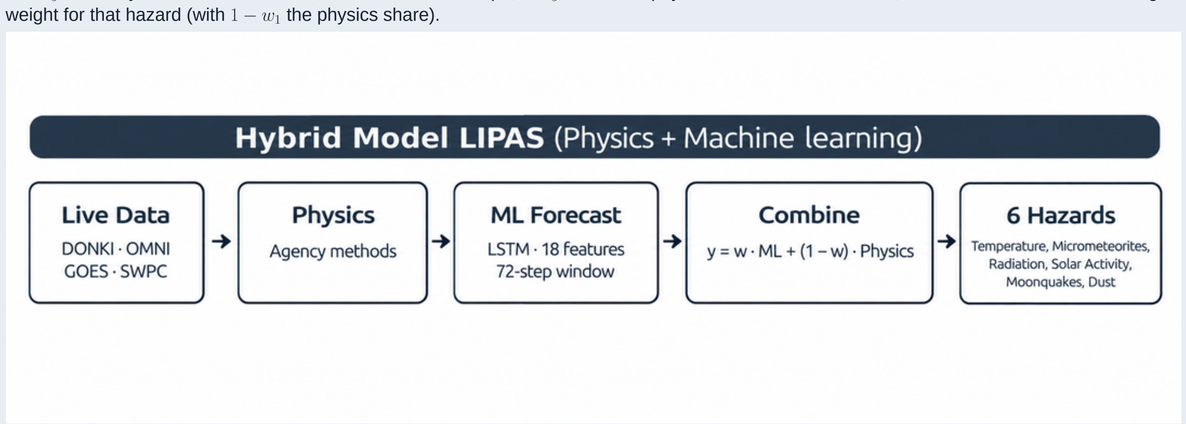
Site map
Inspect any lunar location with local hybrid forecasts.

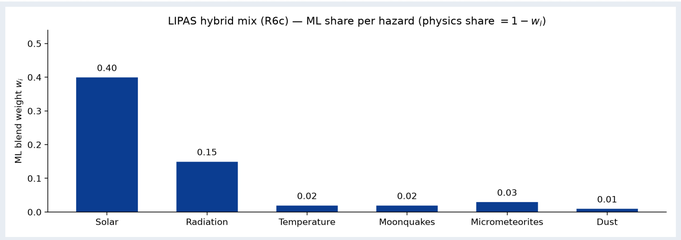
Methods

LIPAS forecasts each lunar hazard by joining physics based estimates (agency style NASA and NOAA methods, live APIs, and historical archives) with a temporal LSTM that reads 18 features over a 72 day window. The two streams are blended with locked, hazard specific weights so machine learning corrects channels where it helps and physics leads where the relation is already strong. The formula below states that blend; the flowchart shows the live data path into six hazard outputs. The experimental design used a qualitative ranking against single purpose tools; a quantitative evaluation at three locations (equatorial reference, Apollo 11, and Nobile Rim) across two locked windows (summer Window A, 1 June to 4 August 2026, 195 site days of quiet to mixed space weather; storm Window B, 20 April to 20 May 2024, 93 site days of disturbed drivers) so accuracy could be judged in both quiet and storm regimes; a radiation focused EVA mission simulation for warning time; and a cross hazard input ablation.

$$\hat{y}_t = w_1\, ML_t + (1 - w_1)\, Physics_t$$

Here $\hat{y}_t$ is the hybrid forecast at time t, ML_t is the LSTM output, $Physics_t$ is the physics based estimate, and w_1 is the locked machine learning weight for that hazard (with $1 - w_1$ the physics share).



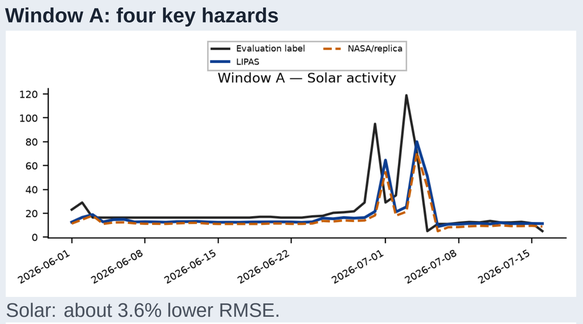


Locked hybrid mix: solar and radiation keep learned corrections; temperature, moonquakes, micrometeorites, and dust stay mostly physics led.

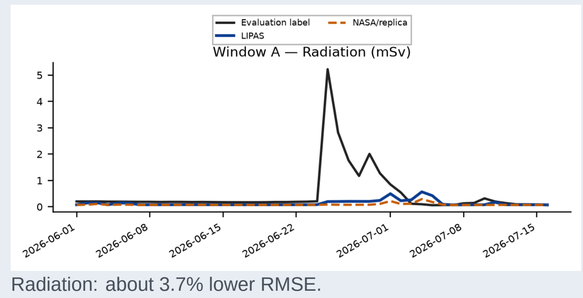
Results

Across both locked evaluation windows, LIPAS beat the replicated baselines on every hazard while temperature, moonquakes, and dust stayed nearly physics tied. Summer Window A (1 June to 4 August 2026, 195 site days) cut solar RMSE by about 3.6% and radiation by about 3.7%. Storm Window B (20 April to 20 May 2024, 93 site days) was the standout: solar improved by about 18.6% and radiation by about 58.7%. Micrometeorite gaps looked huge when AMS shower timing was included versus MEM 3 seasonal only, but those labels are circular, so they illustrate shower timing rather than an independent win. In the EVA mission simulation, LIPAS issued caution at T+50, earlier than combined NASA (T+55) and far earlier than GOES only (T+80), and reached no go at T+55 versus GOES T+105, showing the ops value of one unified forecast view.

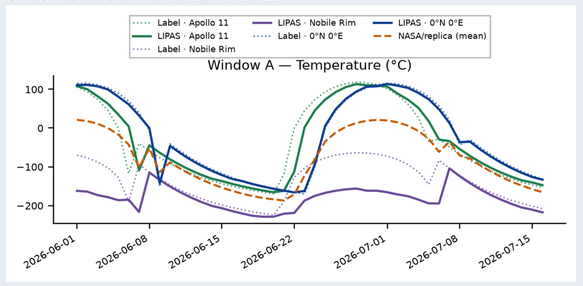
Window A: four key hazards



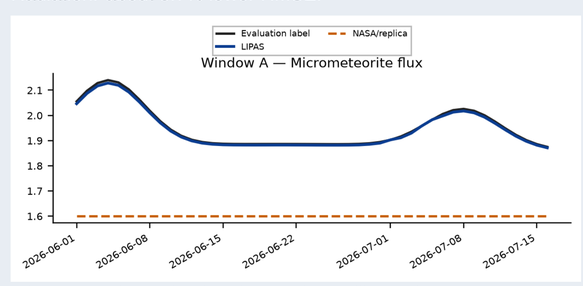
Solar: about 3.6% lower RMSE.



Radiation: about 3.7% lower RMSE.



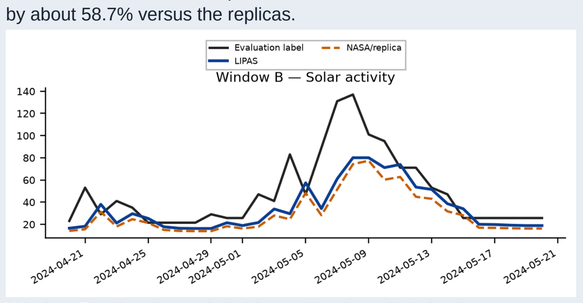
Temperature: three sites where latitude matters.



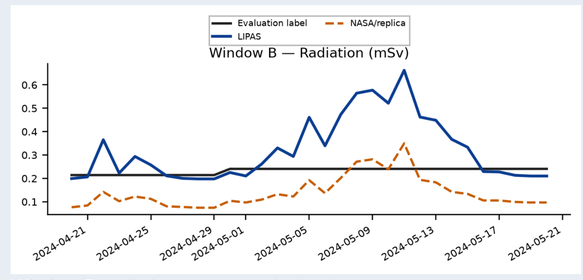
Micrometeorites: circular shower timing illustration.

Window B: storm solar and radiation

Storm Window B stresses space weather drivers. These charts show the clearest hybrid separation: solar RMSE fell by about 18.6% and radiation by about 58.7% versus the replicas.



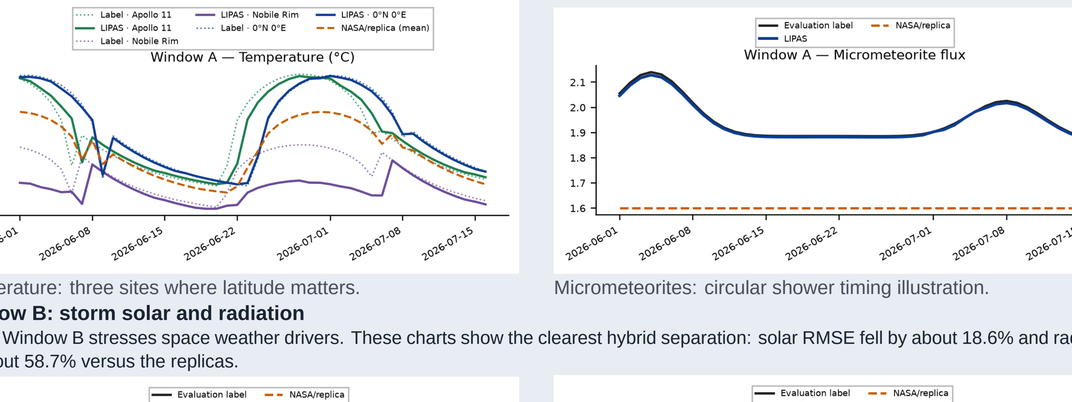
Window B solar: larger hybrid margin when disturbed.



Window B radiation: strongest locked accuracy gain.

Mission simulation

Apollo length EVA with a shared radiation script. LIPAS caution T+50 / no go T+55; combined NASA T+55 / T+70; GOES only T+80 / T+105.



Earlier unified warnings from upstream solar and geomagnetic drivers, not proton flux alone.

Discussion and Conclusion

This study advances a working hybrid lunar forecast stack that joins machine learning with physics in one decision support platform rather than leaving planners to stitch single hazard products by hand.

RQ1. Physics and machine learning can be combined into one lunar forecast system: locked blend weights, a 72 day LSTM trunk, and six hazard outputs were built, served, and evaluated end to end.

RQ2. Unifying hazards improved operational usability. Qualitative ranking favoured the single view for planning, and the EVA simulation showed LIPAS flagging rising radiation earlier (caution T+50, no go T+55) than combined NASA (T+55 / T+70) and GOES only (T+80 / T+105), while specialised NASA tools still lead on scientific depth.

RQ3. Selected accuracy gains versus replicated baselines were real and larger in storm conditions: Window A cut solar and radiation RMSE by about 3.6% and 3.7%; Window B improved them by about 18.6% and 58.7%. Temperature, moonquakes, and dust stayed nearly physics tied. Micrometeorite gaps that use AMS shower timing are circular illustrations, not independent forecast wins.

LIPAS stands as a research grade decision support layer for lunar planners. It should not make crew decisions without human input, and further validation is required before it could serve as a complete mission control system.

References and Acknowledgements

SOURCES USED:
NASA DONKI API: https://api.nasa.gov/DONKI/, NOAA SWPC (GOES protons, X ray, Kp): https://services.swpc.noaa.gov/, NASA OMNI2 Solar Wind Database: https://omniweb.gsfc.nasa.gov, NASA MEM3 Meteoroid Model: https://www.nasa.gov/mem, LADEE LDEX Dust Data: https://pds.nasa.gov/ds-view/pds/viewProfile.jsp?dsid=LADEE-LDEX, Apollo PSE Seismic Catalog: https://pds-geosciences.wustl.edu/missions/apollo/seismic.htm, LRO CRaTER Radiation Products: https://crater-web.sr.unh.edu

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